Sketchbook Upper Bounds

ABSTRACT

CCS CONCEPTS

• **Theory of computation → Bloom filters and hashing**; • **Information systems → Unidimensional range search**.

KEYWORDS

Range Filter, Dynamic Data Structure, Data Growth, Scalability.

ACM Reference Format:

. 2025. Sketchbook Upper Bounds. In *Proceedings of VLDB International Conference on Very Large Databases (2025) (VLDB '25).* ACM, New York, NY, USA, 4 pages. https://doi.org/XXXXXXX.XXXXXXX

1 THEORETICAL ANALYSIS AND RESULTS

For simplicity, we will first discuss the analysis for the case in which we have an insertion-only stream, and later discuss why our arguments carry over when we also have deletions. In the following, we will also assume that, for both CS and CMS, the number of arrays we employ is $d = 1$. This does not change the expected error (in the CMS case) or the variance of the error (in the CS case), merely the confidence we have over the estimate.

We will do the analysis assuming a general size function W and then discuss how the analysis proceeds for specific instantiantions of W (linear or sub-linear regime). Note that we perform doubling each time the function W doubles. In the CMS case, W is a function of $N = \sum f(x)$, the ℓ_1 norm of the stream. In the CS case, W is a function of $F = \sum f^2(x)$, the squared ℓ_2 norm of the stream. In either case, we can track the moments *in time* in which the value of the function doubles. Formally, instead of thinking of W as a function of the the ℓ_1 or ℓ_2 norms of the stream, we think of it as a function of time such that, at any point in time t, W_t is the number of counters the sketch uses at time t.

Division of time into epochs. Let T_0 be the first time the sketch is instantiated: this sketch has W_0 counters and is used as long as the length of the stream is at most N_0. We then define T_i for $i \geq 1$ to be the first moment in time when the length of the array doubles, i.e., $W_i = 2 \cdot W_{i-1}$. Note that, in between times T_i and T_{i+1}, the length of the array stays the same (i.e., it is equal to W_i). We call this period of time epoch i and denote by $\mathcal{S}_i$ the *active* sketch during this epoch (i.e., the one that has arrays of length W_i). According to our design, updates to the stream during Epoch i are executed directly on $\mathcal{S}_i$. For the purposes of executing queries fast, we also aggregate all previous sketches into the active sketch.

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VLDB '25, June 22–27, 2025, London, United Kingdom

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ACM ISBN 978-1-4503-XXXX-X/18/06…$15.00
https://doi.org/XXXXXXX.XXXXXXX

Formally, let $(x, 1)$ be an update that occured in some epoch $0 \leq j < i$ before the current epoch. In the aggregation step, this update operation is replicated on all the counters in $\mathcal{S}_i$ that x could have hashed into. Namely, if x hashed to an index $\mathcal{J}_x$ in epoch j for one of the arrays, then, during epoch i, all of the 2^{j-1} counters $\{0,1\}^{j-i} \circ \mathcal{J}_x$ ingest the update (x, Δ), where $\circ$ denotes concatenation and $\{0,1\}^{j-i}$ denotes all binary strings of length $j-1$. From the perspective of the fixed counter, what this means is that the counter ingests previous updates from elements that don't technically hash into it. We describe this in more detail in what follows.

Agreggating counters from previous epochs. Let $\mathcal{I}$ be a counter in some epoch i. This counter ingests the following operations:

- updates $(x, 1)$ from epoch i and for which $h(x) = \mathcal{I}$. This event occurs with probability $1/W_i$ since there are W_i counters in the array during that epoch. For ease of analysis, we define $C_{i,\mathcal{I}}(x)$ to be the indicator random variable that is 1 if this event occurs, and 0 otherwise. Then $\Pr\left[C_{i,\mathcal{I}}(x) = 1\right] = 1/W_i$.
- updates $(x, 1)$ from epoch $i-1$ that, during epoch $i-1$, hashed into a counter that matches the rightmost $\log W_i - 1$ bits of $\mathcal{I}$ in its binary representation. This event occurs with probability $2/W_i$. The factor 2 accounts for the mistakes that the aggregation step makes, i.e., processing updates from previous epoch that would not have hashed into $\mathcal{I}$ in the current epoch.
- in general, updates $(x, 1)$ from epoch $j < i$ and for which $h(x)$ matches $\mathcal{I}$ in the rightmost j bits of its binary representation. This event occurs with probability $2^{i-j}/W_i$. Similarly as before, we define $C_{j,\mathcal{I}}(x)$ to be the indicator random variable that is 1 if this event occurs, and 0 otherwise. Then $\Pr\left[C_{j,\mathcal{I}}(x) = 1\right] = 2^{i-j}/W_i$.

While updates from previous epochs have a higher chance of hashing into $\mathcal{I}$ than updates in epoch i, we can nevertheless bound the influence they have on the total error of the estimate returned by CMS, CS and Misra Gries.

Streams with deletions. In order to account for deletions, we note that the state of Sketchbook in a stream with deletions is dominated by the state of Sketchbook on a related insertion-only stream. To begin with, notice that, due to the way we perform contractions, we can pretend that all previous epochs (except the one we're actively in), have only insertions in them (the state of counters in Sketchbook is exactly the same as if we only had insertions previously). We now argue about deletions $(x, -1)$ that only occur during some active epoch i and that have not triggered a contraction. If, during the same epoch i, there is also an insertion $(x, 1)$, then we can act as if this (insertion, deletion) pair did not occur in the stream, i.e., the value of the counter incurs no change during that epoch i. Any extra deletions during epoch i that are not cancelled out by insertions during the same epoch can only decrement the value of the counter, and they do not negatively affect the upper bounds we compute. In other words, the value of the counter in epoch i is smaller than its value in a stream in which we remove these extra deletions. Most

crucially, by considering this insertions-only stream, we do not miss any errors the counter accumulates from previous epochs. Namely, during epoch i might still incur errors from previous insertions of x from epochs $j < i$ that would not have hashed to the counter during epoch i (i.e., the event $C_{j,\mathcal{I}}(x)$ still holds and we cannot undo it since we do not know when the last insertion of x occured).

1.1 Bounds for CMS

In the case of CMS, each counter gathers the sum of frequencies of elements that hash into it and returns this as the estimate. We show the following:

Lemma 1.1. *Let $E_W(N)$ denote the estimate of Sketchbook$_{CMS}$ with size function W and one array of counters, on a stream of ℓ_1 norm N. Then, at the end of epoch i, we have the following:*

$$\mathsf{E}\left[E_W(N_{i+1})\right] \leq \frac{1}{W_i} \cdot \sum_{j=0}^{i} 2^{i-j} \cdot (N_{j+1} - N_j)\,,$$

where N_j for $j \geq 0$ denotes the ℓ_1 norm of the stream at the start of epoch j.

Proof. During epoch i, we have $N_{i+1} - N_i$ fresh insertions (i.e., that have not been cancelled out by deletions). We group all these insertions that occur during a given epoch into one big insertion per element. Formally, let $f_i(x)$ denote the minimum of 0 and the total count of updates of the form $(x, 1)$ that occur during epoch i minus the updates of the form $(x, -1)$ from that same epoch . Note that $\sum_x f_j(x) = N_{j+1} - N_j$ for all $j \geq 0$. Note that, in our previous notation, $E_W(N_{i+1}) = \mathcal{I}$ and can be expressed as:

$$\mathcal{I} = \sum_{j=0}^{i} \sum_{x} C_{j,\mathcal{I}}(x) \cdot f_j(x)\,.$$

In other words, the counter $\mathcal{I}$ gets the frequency $f_j(x)$ added to it if and only if the event $C_{j,\mathcal{I}}(x)$ occurs. We also have that $\mathsf{E}\left[C_{j,\mathcal{I}}(x)\right] \leq 2^{i-j}/W_i$. By linearity of expectation, we then get that:

$$\begin{aligned} E[\mathcal{I}] &\leq \sum_{j=0}^{i} \sum_{x} \frac{2^{i-j}}{W_i} \cdot f_j(x) = \frac{1}{W_i} \cdot \sum_{j=0}^{i} \sum_{x} 2^{i-j} \cdot f_j(x) \\ &= \frac{1}{W_i} \cdot \sum_{j=0}^{i} 2^{i-j} \sum_{x} f_j(x) = \frac{1}{W_i} \cdot \sum_{j=0}^{i} 2^{i-j} \cdot (N_{j+1} - N_j)\,. \end{aligned}$$

The claim follows. □

In other words, the increase in error we get because of aggregating the sketches is dominated by the term $\sum_{j=0}^{i} 2^{i-j} \cdot (N_{j+1} - N_j)$. This is where the specific size function W we employ has an effect, since it translates into bounds on N_j (and specifically their relative sizes). We now consider various special cases and discuss their respective behavior:

Theorem 1.2. *Let $E_W(N)$ denote the estimate of Sketchbook$_{CMS}$ with size function W and one array of counters, on a stream of ℓ_1 norm $N = \sum_y f(y)$. Then:*

$$\mathsf{E}\left[E_W(N)\right] \leq \begin{cases} \log_2(N) \cdot \frac{N}{W(N)}, & \text{if } W(N) = N/\epsilon, \\ \frac{N}{W(N)}, & \text{if } W(N) = N^{1-\alpha}/\epsilon, \text{ for } 0 < \alpha < 1 \end{cases}.$$

Proof. In the case in which $W(N) = N/\epsilon$, the size function doubles whenever the number of elements doubles, i.e., $N_{j+1} = 2 \cdot N_j$ for all $j \geq 0$ and, in general, $N_j = 2^j \cdot N_0$ for all $j \geq 0$. The additional error term then becomes:

$$\begin{aligned} \sum_{j=0}^{i} 2^{i-j} \cdot (N_{j+1} - N_j) &= \sum_{j=0}^{i} 2^{i-j} \cdot (2^{j+1} \cdot N_0 - 2^j \cdot N_0) \\ &= \sum_{j=0}^{i} 2^{i-j} \cdot 2^j \cdot N_0 = (i+1) \cdot 2^i \cdot N_0 \\ &= (i+1) \cdot N_i\,. \end{aligned}$$

Plugging in this bound in the formula from Lemma 1.1 (i.e., dividing by $W(N_i) = W_i$), we get the main claim for $W(N) = N/\epsilon$.

The case in which $W(N) = N^{1-\alpha}/\epsilon$, for some $0 < \alpha < 1$. In this case, the size function doubles more rarely, whenever $N_{i+1} = 2^{1/(1-\alpha)} \cdot N_i$. Since $0 < \alpha < 1$, note that this implies that N_{i+1} is bigger than $2 \cdot N_i$. This difference is why we avoid the extra dependency on i in the error term. Formally, we have that $N_i = 2^{i/(1-\alpha)} \cdot N_0$ and thus:

$$\begin{aligned} \sum_{j=0}^{i} 2^{i-j} \cdot (N_{j+1} - N_j) &= \sum_{j=0}^{i} 2^{i-j} \cdot (2^{(j+1)/(1-\alpha)} \cdot N_0 - 2^{j/(1-\alpha)} \cdot N_0) \\ &= \sum_{j=0}^{i} 2^{i-j} \cdot 2^{j/(1-\alpha)} \cdot N_0 \cdot (2^{1/(1-\alpha)} - 1) \\ &= 2^i \cdot N_0 \cdot (2^{1/(1-\alpha)} - 1) \cdot \sum_{j=0}^{i} 2^{j/(1-\alpha)-j}\,. \end{aligned}$$

For simplicity, we define $2^{1/(1-\alpha)-1} = a$ and observe that:

$$\sum_{j=0}^{i} 2^{j/(1-\alpha)-j} = \sum_{j=0}^{i} a^j = (a^{i+1} - 1)/(a-1)\,.$$

We then get that:

$$\begin{aligned} \sum_{j=0}^{i} 2^{i-j} \cdot (N_{j+1} - N_j) &= 2^i \cdot N_0 \cdot (2^{1/(1-\alpha)} - 1) \cdot \sum_{j=0}^{i} 2^{j/(1-\alpha)-j} \\ &= 2^i \cdot N_0 \cdot (2a-1) \cdot (a^{i+1} - 1)/(a-1) \\ &\leq 2^{i+1} \cdot N_0 \cdot (a^{i+1} - 1) \leq 2^{i+1} \cdot N_0 \cdot a^{i+1}\,. \end{aligned}$$

We now notice that $2a = 2^{1/(1-\alpha)}$ and so

$$2^{i+1} \cdot N_0 \cdot a^{i+1} = 2^{(i+1)/(1-\alpha)} \cdot N_0 = N_{i+1}\,.$$

Therefore, the expected error from Lemma 1.1 becomes:

$$E[\mathcal{I}] \leq \frac{1}{W_i} \cdot \sum_{j=0}^{i} 2^{i-j} \cdot (N_{j+1} - N_j) \leq \frac{N_{i+1}}{W_i}\,.$$

The claim follows.

□

1.2 Bounds for CS

In the case of CS, we obtain a similar claim to that of Lemma 1.1, except as a function of the squared ℓ_2 norm of the stream.

Lemma 1.3. *Let $V_W(F)$ denote the estimate of Sketchbook$_{CS}$ with size function W and one array of counters, on a stream with squared ℓ_2 norm $F = \sum_y f(y)^2$. Then, at the end of epoch i, we have the following:*

$$Var\left[V_W(F_{i+1})\right] \leq \frac{1}{W_i} \cdot \sum_{j=0}^{i} 2^{i-j} \cdot (F_{j+1} - F_j)\,,$$

where F_j denotes the squared ℓ_2 norm of the stream at the start of epoch j.

Proof. We are interested in describing the behavior of the random variable $\hat{f}(z) = \mathcal{I} \cdot \sigma(z)$ for any query z, where $\sigma(z)$ is the associated sign function (for simplicity, we again assume that $d = 1$) and $\mathcal{I}$ is the counter that z hashes into. Using the notation we introduced before, this counter $\mathcal{I}$ can be expressed as:

$$\mathcal{I} = \sum_{j=0}^{i} \sum_{x} C_{j,\mathcal{I}}(x) \cdot f_j(x) \cdot \sigma(x) = \sum_{x} C(x)\,,$$

where $C(x) = \sum_{j=0}^{i} C_{j,\mathcal{I}}(x) \cdot f_j(x) \cdot \sigma(x)$ denotes the contribution to counter $\mathcal{I}$, throughout the epochs, of updates to x.

We begin by computing the expected value of the estimator $\hat{f}(z)$ as in the classical CS analysis that shows that in expectation, $\mathcal{I} \cdot \sigma(z)$ is an unbiased estimator, i.e. $E[\mathcal{I} \cdot \sigma(z)] = f(z)$. In the following, we let $f_j(x)$ denote the frequency of x when restricted to epoch j. By linearity of expectation

$$\mathsf{E}[\hat{f}(z)] = \sum_{x \neq z} \mathsf{E}\left[C(x) \cdot \sigma(z)\right] + \mathsf{E}\left[C(z)\sigma(z)\right] = \sum_{j=0}^{i} f_j(z) = f(z)\,,$$

where the last step uses the fact that $\mathsf{E}\left[\sigma(x) \cdot \sigma(z)\right] = 0$ for any $x \neq z$ and that $\Pr\left[C_{j,\mathcal{I}}(z)\right] = 1$ since $h(z) = \mathcal{I}$. We will now bound the variance of this estimator, and in particular, we will be interested in how much bigger it becomes when compared to $\sqrt{F_i/W_i}$. We get that:

$$\begin{aligned}
\mathsf{Var}\left[\hat{f}(z)\right] &= \mathsf{E}\left[\left(\sum_x C(x)\sigma(z)\right)^2\right] - (f(z))^2 \\
&= \sum_x \mathsf{E}\left[(C(x))^2\right] + \sum_x \sum_{y \neq x} \mathsf{E}\left[C(x)C(y)\right] - (f(z))^2 \\
&= \sum_x \mathsf{E}\left[(C(x))^2\right] - (f(z))^2\,,
\end{aligned}$$

where we use the fact that $\mathsf{E}\left[\sigma(x) \cdot \sigma(y)\right] = 0$ for any $x \neq y$ to establish that $\mathsf{E}\left[C(x)C(y)\right] = 0$. We now bound $\mathsf{E}\left[(C(x))^2\right]$. Recall that we defined $C(x) = \sum_{j=0}^{i} C_{j,\mathcal{I}}(x) \cdot f_j(x) \cdot \sigma(x)$. Then:

$$\begin{aligned}
\mathsf{E}\left[(C(x))^2\right] &= \mathsf{E}\left[\left(\sum_{j=0}^{i} C_{j,\mathcal{I}}(x) \cdot f_j(x)\right)^2\right] \\
&= \sum_{j=0}^{i} \mathsf{E}\left[\left(C_{j,\mathcal{I}}(x) \cdot f_j(x)\right)^2\right] + \\
&+ 2 \cdot \sum_{j=0}^{i-1} \sum_{k=j+1}^{i} \mathsf{E}\left[C_{j,\mathcal{I}}(x) \cdot f_j(x) \cdot C_{k,\mathcal{I}}(x) \cdot f_k(x)\right] \\
&= \sum_{j=0}^{i} f_j(x)^2 \cdot \mathsf{E}\left[C_{j,\mathcal{I}}(x)\right] + \\
&+ 2 \cdot \sum_{j=0}^{i-1} \sum_{k=j+1}^{i} f_j(x) \cdot f_k(x) \cdot \mathsf{E}\left[C_{j,\mathcal{I}}(x) \cdot C_{k,\mathcal{I}}(x)\right]\,.
\end{aligned}$$

For bounding the $\mathsf{E}\left[C_{j,\mathcal{I}}(x)\right]$, we can use the fact that $\mathsf{E}\left[C_{j,\mathcal{I}}(x)\right] \leq 2^{i-j}/W_i$. For $\mathsf{E}\left[C_{j,\mathcal{I}}(x) \cdot C_{k,\mathcal{I}}(x)\right]$, we note that the product of the two indicator random variables is 1 only when both of them are 1. Having said that, the two random variables are not independent: if $C_{k,\mathcal{I}}(x) = 1$, then $C_{j,\mathcal{I}}(x) = 1$ for any $k \geq j$. Namely, if during epoch k, x hashed into a counter that was a prefix of $\mathcal{I}$, then the same would have happened during epoch $j \leq k$. We get that:

$$\mathsf{E}\left[C_{j,\mathcal{I}}(x) \cdot C_{k,\mathcal{I}}(x)\right] = \mathsf{E}\left[C_{k,\mathcal{I}}(x)\right]\,.$$

The whole sums becomes:

$$\begin{aligned}
\sum_{j=0}^{i-1} \sum_{k=j+1}^{i} f_j(x) \cdot f_k(x) \cdot \mathsf{E}\left[C_{k,\mathcal{I}}(x)\right] &= \sum_{k=1}^{i} \sum_{j=0}^{k-1} f_j(x) \cdot f_k(x) \cdot \mathsf{E}\left[C_{k,\mathcal{I}}(x)\right] \\
&= \sum_{k=1}^{i} f_k(x) \cdot \mathsf{E}\left[C_{k,\mathcal{I}}(x)\right] \cdot \sum_{j=0}^{k-1} f_j(x) \\
&= \sum_{k=1}^{i} f_k(x) \cdot \mathsf{E}\left[C_{k,\mathcal{I}}(x)\right] \cdot f_{\leq k-1}(x)
\end{aligned}$$

where, for simplicity, we denoted $f_{\leq i}(x) = \sum_{j=0}^{i} f_i(x)$ the sum of frequencies accumulated by x through epochs 0 to i and set $f_{\leq -1}(x) = 0$. Then:

$$\begin{aligned}
\mathsf{E}\left[(C(x))^2\right] &= \sum_{j=0}^{i} \left(f_j(x)^2 + 2 f_j(x) \cdot f_{\leq j-1}(x)\right) \cdot \mathsf{E}\left[C_{j,\mathcal{I}}(x)\right] \\
&= \sum_{j=0}^{i} \left(f_{\leq j}(x)^2 - f_{\leq j-1}(x)^2\right) \cdot \mathsf{E}\left[C_{j,\mathcal{I}}(x)\right]\,.
\end{aligned}$$

While the term $f_{\leq j}(x)^2 - f_{\leq j-1}(x)^2$ might seem curious, we note that $\sum_x f_{\leq j}(x)^2 - f_{\leq j-1}(x)^2 = F_j - F_{j-1}$ denotes the increase in the squared ℓ_2 norm of the stream vector as we move from epoch $j-1$ to epoch j. We thus get that:

$$\begin{aligned}\mathrm{Var}\left[\hat{f}(z)\right] &= \sum_x \mathsf{E}\left[(C(x))^2\right] - (f(z))^2 \\ &= \sum_x \sum_{j=0}^{i} \left(f_{\leq j}(x)^2 - f_{\leq j-1}(x)^2\right) \cdot \mathsf{E}\left[C_{j,\mathcal{I}}(x)\right] - (f(z))^2 \\ &\leq \sum_{j=0}^{i} \left(F_j - F_{j-1}\right) \cdot \frac{2^{i-j}}{W_i} \ .\end{aligned}$$

The claim follows. □

At this point, it becomes perhaps more natural to understand why we defined the size function to depend on the squared ℓ_2 norm of the stream vector. We now discuss different bounds we get on the variance depending on the size function.

Theorem 1.4. *Let $V_W(F)$ denote the estimate of Sketchbook$_{CS}$ with size function W and one array of counters, on a stream with squared ℓ_2 norm $F = \sum_y f(y)^2$. Then:*

$$Var\left[V_W(F)\right] \leq \begin{cases} \log_2(F) \cdot \frac{F}{W(F)}, & \text{if } W(F) = F/\epsilon, \\ \frac{F}{W(F)}, & \text{if } W(F) = F^{1-\alpha}/\epsilon, \text{ for } 0 < \alpha < 1 \end{cases} .$$

We note that the calculations from the above are highly similar to those of Sketchbook$_{\mathrm{CMS}}$ from Theorem 1.2, with the difference that we now see a dependency on F rather than N. We include them for completeness:

Proof. In the case in which $W(F) = F/\epsilon$, we start with the original sketch $\mathcal{S}_0$ that allows streams for which their squared ℓ_2 norm is at most some F_0. The size function doubles whenever the squared ℓ_2 norm doubles. In this case $F_{j+1} = 2 \cdot F_j$ for all $j \geq 0$ and, in general, $F_j = 2^j \cdot F_0$ for all $j \geq 0$. In this case, we get that:

$$\begin{aligned}\mathrm{Var}\left[V_W\right] &\leq \sum_{j=0}^{i} \left(F_j - F_{j-1}\right) \cdot \frac{2^{i-j}}{W_i} \\ &\leq \frac{1}{W_i} \cdot \left(F_0 \cdot 2^i + \sum_{j=1}^{i} \left(2^j - 2^{j-1}\right) \cdot F_0 \cdot 2^{i-j}\right) \\ &\leq \frac{1}{W_i} \cdot \left(F_0 \cdot 2^i + \sum_{j=1}^{i} 2^{j-1} \cdot F_0 \cdot 2^{i-j}\right) \\ &\leq \frac{1}{W_i} \cdot \left(F_0 \cdot 2^i + \sum_{j=1}^{i} F_0 \cdot 2^{i-1}\right) \\ &\leq \frac{1}{W_i} \cdot (F_i + (i-1) \cdot F_i/2) = \frac{F_i}{W_i} \cdot \frac{i+1}{2} \ .\end{aligned}$$

Setting $W(F) = F^{1-\alpha}/\epsilon$ in CS, for some $0 < c < 1$. In this case, the size function doubles more rarely, whenever $F_{i+1} = 2^{1/(1-\alpha)} \cdot F_i$. Since $0 < \alpha < 1$, note that this implies that F_{i+1} is bigger than $2F_i$. This difference is why we avoid the extra dependency on i in the error term. Plugging this into the statement of Lemma 1.3, we have that $F_i = 2^{i/(1-\alpha)} \cdot F_0$ and thus:

$$\begin{aligned}\sum_{j=0}^{i} 2^{i-j} \cdot (F_{j+1} - F_j) &= \sum_{j=0}^{i} 2^{i-j} \cdot (2^{(j+1)/(1-\alpha)} \cdot T_0 - 2^{j/(1-\alpha)} \cdot T_0) \\ &= \sum_{j=0}^{i} 2^{i-j} \cdot 2^{j/(1-\alpha)} \cdot T_0 \cdot (2^{1/(1-\alpha)} - 1) \\ &= 2^i \cdot T_0 \cdot (2^{1/(1-\alpha)} - 1) \cdot \sum_{j=0}^{i} 2^{j/(1-\alpha)-j} \ .\end{aligned}$$

□

Received 16 April 2024; revised 4 June 2024; accepted 8 August 2024